

Used cars sale price prediction

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1. Motivation

The used-car market is characterized by a highly diverse range of vehicles, inconsistent listing quality, and complex relationships among vehicle age, mileage, manufacturer, model, body type, fuel type, transmission, and location. As a result, manually estimating a realistic selling price can be time-consuming. The aim of this project is to develop a regression model that estimates a vehicle's selling price based on the available listing attributes. Such a model can support buyers in comparing offers, help sellers set appropriate prices, and assist online platforms in automatically detecting unusually expensive or inexpensive listings.

2. Research questions

The study addresses the following research questions: How accurately can the selling price be estimated from a combination of numerical and categorical attributes? Do engineered features, particularly vehicle age and the ratio between mileage and age, improve the modelling of depreciation? Which regression algorithm achieves the lowest relative error? In which price segments does the model make the most frequent errors?

A used-car dataset obtained from the Kaggle platform was used. After data cleaning, 4,999 listings were retained. The target variable is `sale_price`. The mean price is 454,969, the median is 382,499, and the maximum reaches 3,866,000, indicating a pronounced right-skewed distribution. The average vehicle age is 6.92 years, while the average mileage is 62,754 km. The highest proportions of missing values are found in `car_availability` (8.14%) and `transmission` (7.20%), whereas the remaining columns are considerably more complete.

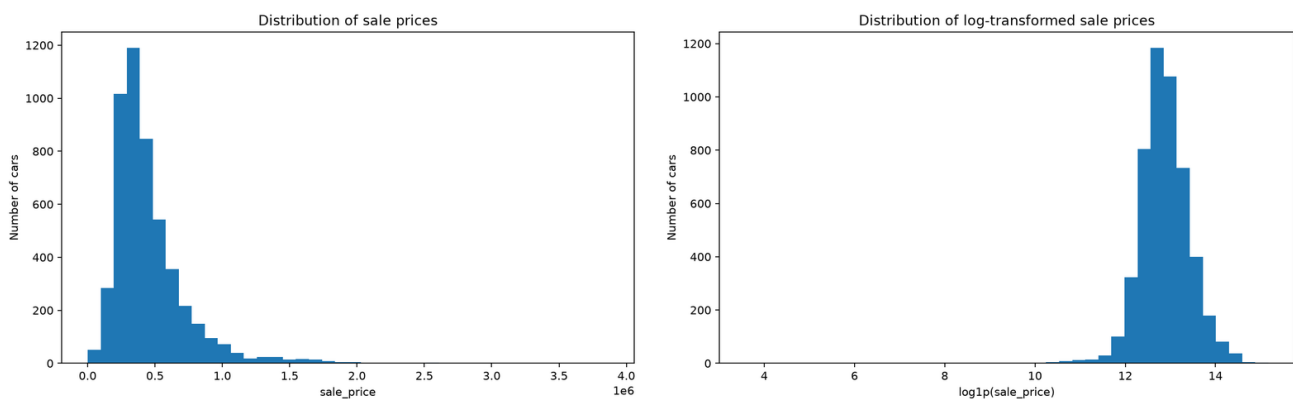


Figure 1. Distribution of the original and log-transformed sale price. The log transformation reduces the right skewness of the target variable.

3. Related work

Vehicle price prediction is a typical regression problem involving nonlinear relationships and a large number of categorical values. Linear and regularized regression provide important baseline models, but on their own they struggle to capture thresholds and interactions among vehicle age, mileage, brand, and vehicle type. Tree-based ensembles are therefore a natural choice: Random Forest combines a large number of decorrelated decision trees, while Gradient Boosting sequentially corrects the errors made by previous weak learners. Stacked Generalization combines the outputs of several different models through a final meta-model and can reduce generalization error. In this study, these groups of methods were systematically compared within a unified preprocessing pipeline implemented using the scikit-learn library.

4. Methodology

Column names were standardized, completely empty columns and duplicate records were removed, and the price was parsed from its textual representation and restricted to positive values. Listing dates were converted into year and month features. Text-based numerical values, such as mileage or year of manufacture, were converted when at least 60% of the column could be parsed reliably. Binary values were mapped to 0/1, while the combined categorical features `make_model`, `make_body_type`, and `fuel_transmission` were created by merging related categories.

Step	Application
Age and usage	<code>car_age</code> , <code>km_per_year</code> , <code>log_kms_run</code> , <code>log_km_per_year</code> , <code>age_km_interaction</code>
Age nonlinearity	<code>car_age_squared</code> , <code>is_newer_car</code> (≤ 3), <code>is_old_car</code> (≥ 10)
Additional features	<code>high_mileage</code> , <code>log_times_viewed</code> , <code>owners_per_year</code>
Numerical pipeline	Median imputation, 1st/99th percentile clipping, StandardScaler
Categorical pipeline	Imputation with unknown, OneHotEncoder. Rare and previously unseen categories supported
Target variable	<code>log1p</code> transformation during training and <code>expm1</code> when converting predictions back to the original scale

The data were stratified into ten price quantiles and split into 80% training and 20% test sets (`random_state=42`). The test set contains 1,000 examples. After the split, observations outside the 5th and 95th price percentiles were removed from the training set, leaving 3,599 listings for model training; the test set remained unchanged. All imputations, outlier thresholds, scaling parameters, and encoding rules were learned exclusively from the training data, thereby preventing information leakage.

The following models were compared: Linear Regression, Ridge Regression, Lasso Regression, Decision Tree, Random Forest, Gradient Boosting, Histogram Gradient Boosting, Extra Trees, Voting Regressor, and Stacking Regressor. The stacking model uses Ridge Regression, Random Forest, Gradient Boosting, and Extra Trees as base learners, three-fold cross-validation, and Ridge Regression as the meta-model. MAPE was used as the primary model-selection metric, while MAE, RMSE, and R^2 served as supplementary evaluation measures. MAPE is suitable for interpreting the average relative error, whereas RMSE penalizes large prediction errors more heavily, particularly for expensive vehicles.

5. Discussion

Table 1 presents the results obtained on the held-out test set. The Stacking Regressor achieved the lowest MAPE (10.62%) and was therefore selected as the final solution. Linear Regression obtained a slightly higher MAPE (10.69%), but achieved better MAE, RMSE, and R^2 scores. This shows that model selection depends on the chosen evaluation criterion: stacking provides the lowest average relative error, while the linear model handles several very large absolute errors more effectively.

Model	MAE	RMSE	R^2	MAPE
Stacking Regressor	52.939	134.077	0,748	10,62%
Linear Regression	50.517	111.650	0,825	10,69%
Ridge Regression	55.892	135.229	0,743	11,17%
Lasso Regression	59.851	140.382	0,724	12,26%
Histogram Gradient Boosting	60.419	149.549	0,686	12,95%
Extra Trees	63.213	153.604	0,669	14,24%
Random Forest	66.089	157.639	0,651	15,13%
Voting Regressor	69.645	159.371	0,644	15,40%
Gradient Boosting	77.480	163.917	0,623	17,18%
Decision Tree	77.074	164.974	0,618	17,40%

Table 1. Results of all evaluated models on the test set; the selected model is highlighted in green.

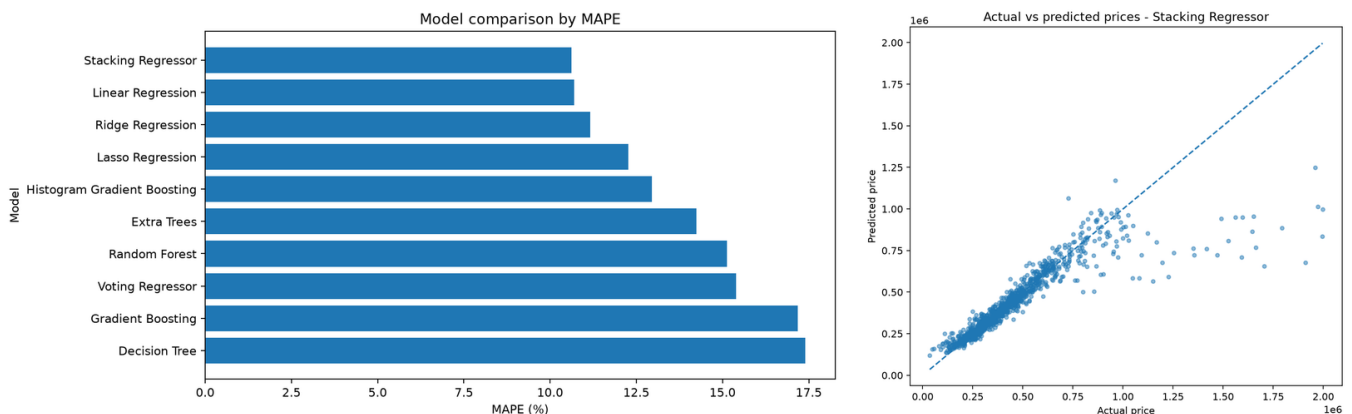


Figure 2. Comparison of the models based on MAPE and the relationship between actual and predicted prices for the Stacking Regressor.

In the actual-versus-predicted price plot, most points within the central price range follow the ideal diagonal closely. Deviations increase for the most expensive vehicles, where predictions are often lower than the actual prices. Overall, 69.2% of the test examples have an error of up to 10%, 83.0% up to 15%, and 89.6% up to 20%. This confirms that the model is useful for typical listings, but less reliable at the extremes of the price distribution.

The lowest price decile has an average MAPE of 29.32%: the mean actual price is 165,587, while the mean predicted price is 196,645, indicating that the model overestimates very inexpensive vehicles. In the highest decile, the mean actual price is 1,053,731, whereas the mean predicted price is 816,579; the MAPE is 20.77%, showing pronounced underestimation. The middle eight deciles have a considerably more stable MAPE, generally ranging between 5.6% and 8.7%. This pattern is consistent with regression toward the mean and with the fact that the most extreme 10% of prices were removed only from the training set. The largest absolute errors occur for rare and expensive models, while the largest percentage errors are found among very inexpensive, older vehicles.

Permutation importance shows that the model depends most strongly on body type (body_type), followed by year of manufacture, brand, squared vehicle age, vehicle age itself, and the interaction between age and mileage. This supports the usefulness of the engineered features. However, these importance scores should not be interpreted causally: yr_mfr, car_age, and car_age_squared are strongly correlated features and may share explanatory information. The residual plot also shows that error variance increases with price.

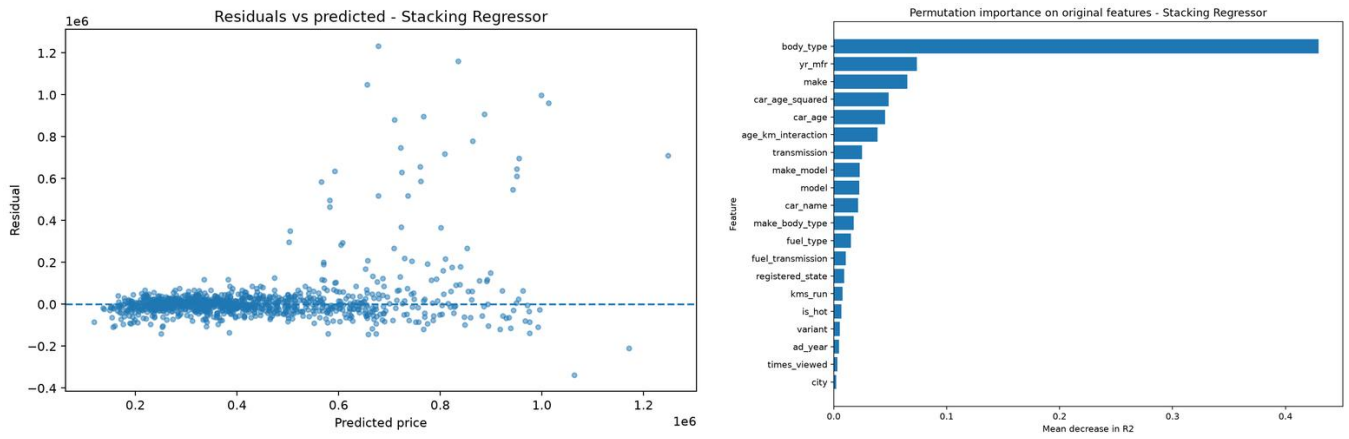


Figure 3. Residuals versus predicted values and permutation importance of the original features

The final solution provides good relative accuracy across the dominant portion of the dataset, but it could be improved by modeling extreme price segments separately, using logarithmic or Huber loss optimized through cross-validation, and including more examples of luxury vehicles. It would also be useful to evaluate the model across multiple repeated train-test splits, since a single test split does not capture the full variability of the results. For practical use, the prediction should be presented as an estimate or interval rather than as an absolutely accurate market price.

6. Reference

- [1] V. Sharma, "Used Car Price Prediction", Kaggle dataset.
- [2] L. Breiman, "Random Forests", Machine Learning, 45(1), 5-32, 2001.
- [3] ChatGPT